GRADE 11 / Mathematics / English

Read a quadratic graph

01 Understand

Learning goals: Find intercepts and the turning point • Connect factors to roots • State a range using the turning point

A quadratic graph has equation $y = ax^2 + bx + c$, with $a \neq 0$. Positive a opens upwards; negative a opens downwards.

The axis of symmetry is $x = -b/(2a)$. Substitute this x-value to find the turning point.

The y-intercept occurs at $x = 0$. The x-intercepts occur where $y = 0$; there may be two, one or no real roots.

Symmetry around x = 2

x	$y = x^2 - 4x + 3$	Meaning
0	3	y-intercept
1	0	root
2	−1	minimum
3	0	root
4	3	mirror of x = 0

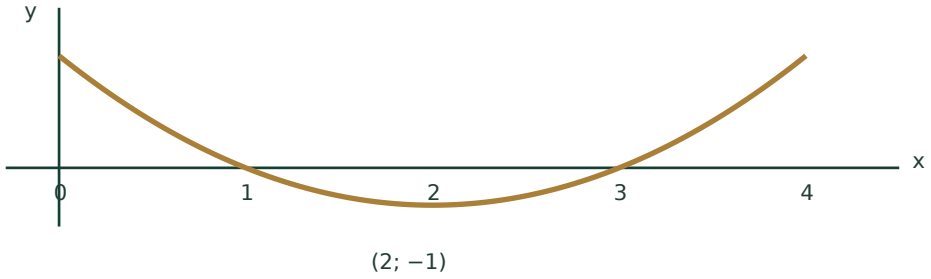
02 Follow the reasoning

Sketch $y = x^2 - 4x + 3$ and state its range.

1. Factorise: $y = (x - 1)(x - 3)$, giving roots 1 and 3.

2. The axis is $x = 2$; $y(2) = -1$. Mark $(2; -1)$ and the y-intercept $(0; 3)$.

3. Draw a smooth upward parabola through these points. The range is $y \geq -1$.



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Practise with purpose

03 Try before checking

Use paper or your notebook. Show your reasoning. For written responses, use evidence and explain your choices. Suggested time: 15–20 minutes.

1. Find the roots of $y = x^2 - 6x + 8$.

2. Find its turning point.

3. State its axis of symmetry and range.

04 Check and correct

1. $(x - 2)(x - 4)$ gives $x = 2$ and $x = 4$.

2. $x = 3$; $y = 9 - 18 + 8 = -1$. Turning point $(3; -1)$.

3. Axis $x = 3$; range $y \geq -1$.

Common mistake to avoid

The axis of symmetry is an equation $x = k$, not a single point or a y-value.

Reflect and revisit

Identify one correction in your own words. Attempt a similar question tomorrow without looking at the guide. If you are still unsure, take your attempt to a teacher or tutor.

Coverage and sources

CAPS topic focus: Functions: quadratic functions

DBE CAPS, Grades 10–12: Mathematics. Section 3 content and skills. This is a focused supplement, not a complete syllabus or prescribed literature replacement.

Open the official DBE curriculum reference

Content and visuals: original SmartBridge FET. AI-assisted authoring; subject-teacher and language-specialist review pending. Report an issue with the pack ID math-11.